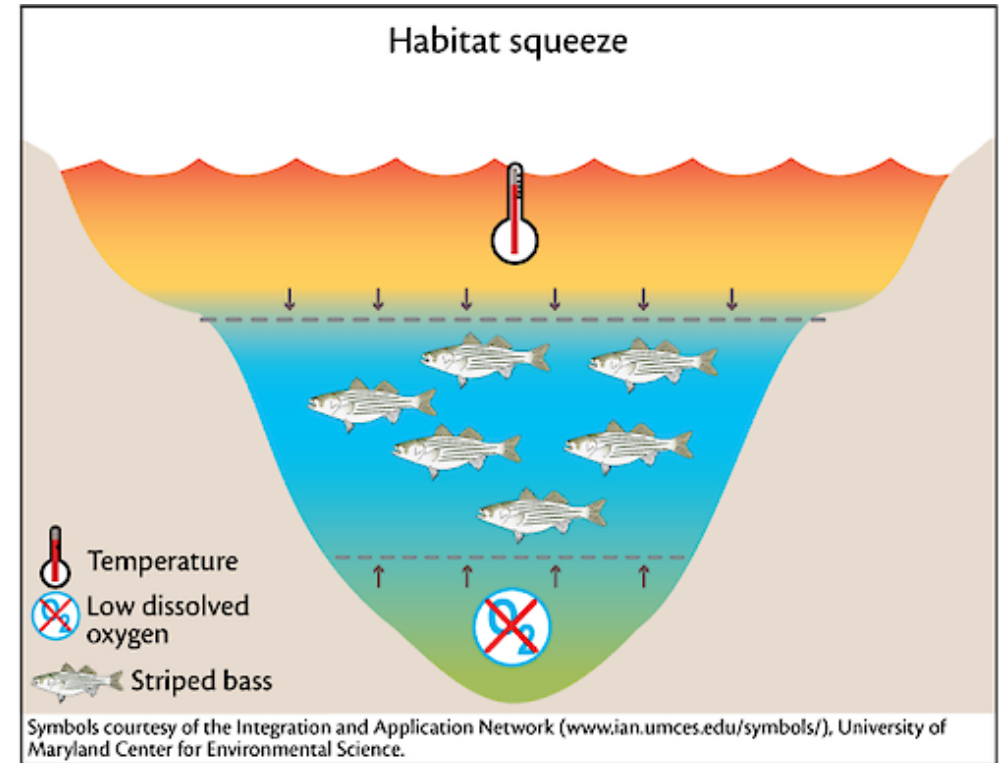
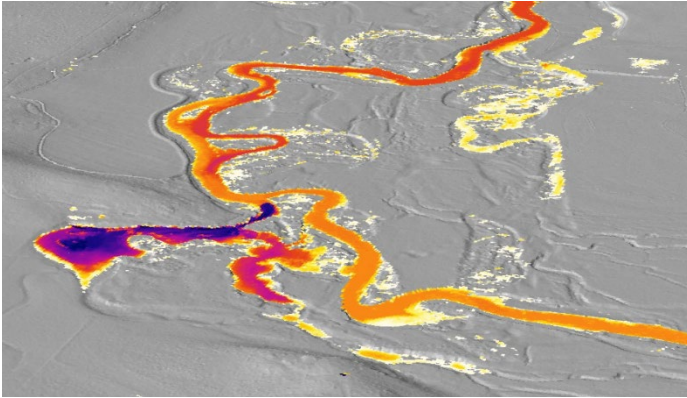


Developing a habitat selection model
to analyze vertical distribution of
coldwater fish in a thermal refuge
habitat

Temperature oxygen squeeze in a coldwater refuge habitat



Conceptual diagram illustrating how change in a water source's oxygen and temperature can restrict the habitat of species.
Diagram courtesy of the Integration and Application Network (ian.umces.edu), University of Maryland Center for Environmental Science. Source: Ecocheck

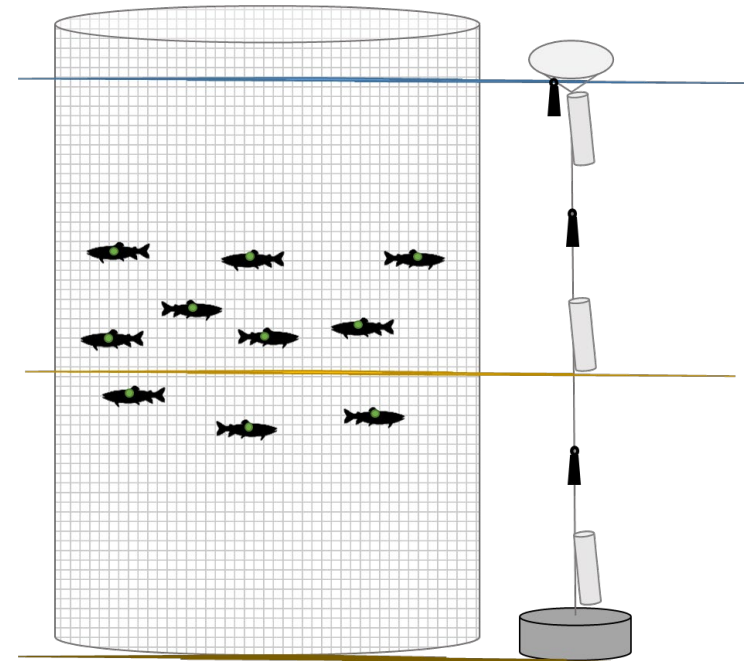
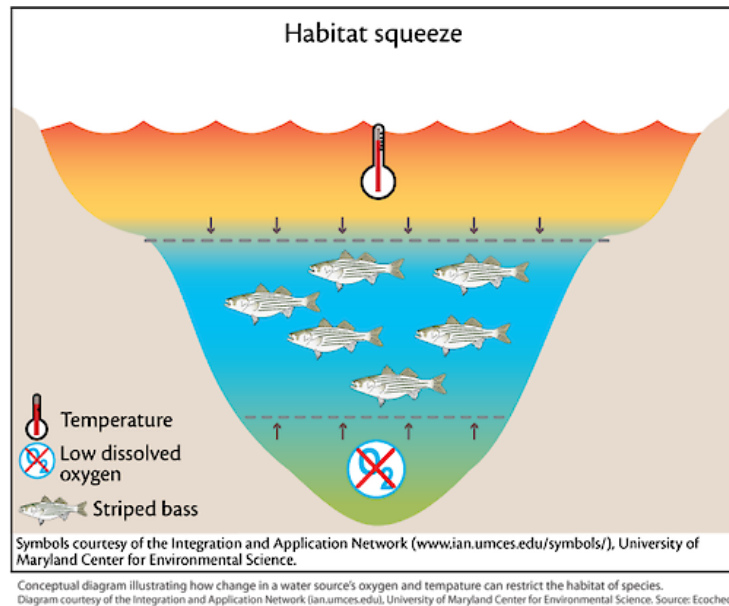
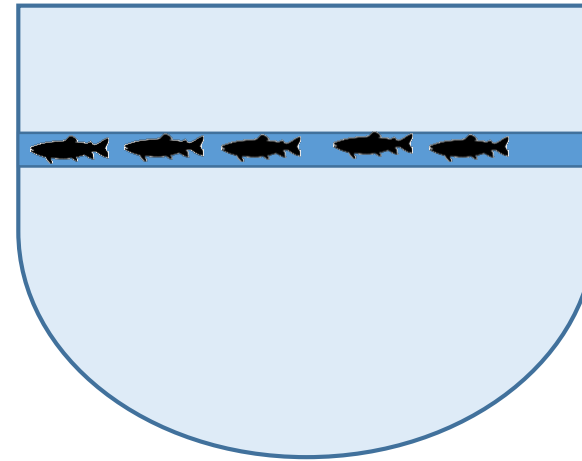
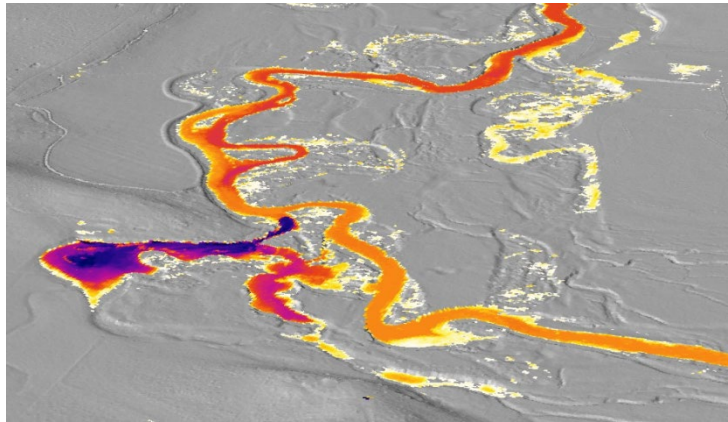
Transactions of the American Fisheries Society 114:51–61, 1985

**Striped Bass, Temperature, and Dissolved Oxygen:
A Speculative Hypothesis for Environmental Risk^{1,2}**

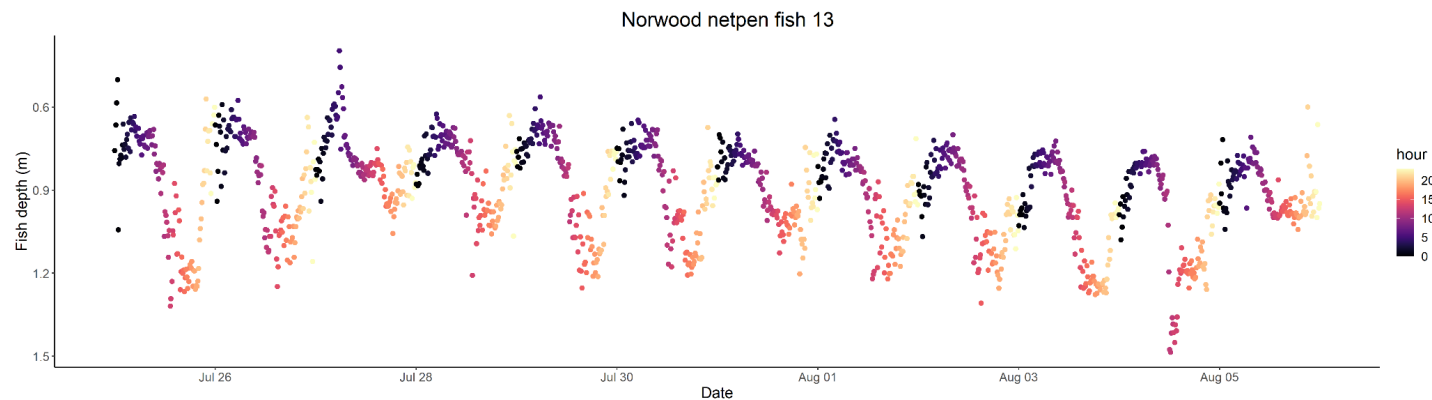
CHARLES C. COUTANT

Temperature oxygen squeeze in a coldwater refuge habitat

Diel vertical movement



- Determine fish depth and dissolved oxygen over time using fish temperature and logger array temperatures recorded at known depths (linear interpolation using slope equation between bounding sensors)
- Generate ~10 random depths (bounded by actual water column depth of study site) using uniform distribution and calculate estimated temp/do per time step using linear interpolation (and known sensor temps and depths)
 - Create a data frame with individual fish 'selected' temperature and dissolved oxygen versus what was available in the water column at that time step
- Step selection function (SSF) to evaluate fish movement in response to fluctuating habitat variables (dissolved oxygen and temperature)



Anticipated challenges

- Must do research on SSF
- How to incorporate time – selection day vs night, does selection change over extent of (~2 week) study period?
- Likely not linear – how to parameterize